Formulation

Problem transformations

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Optimization: principles and algorithms

ÉCOLE POLYTECHNIQUE
FÉDÉRALE DE LAUSANNE

Problem transformations

There are many ways to write an optimization problem.
Algorithms usually require a specific type of formulation.
For instance, most optimization software are designed only for minimization, or maximization.

Equivalence

Definition

Problems P_1 and P_2 are equivalent if a feasible point of P_1 can be created from a feasible point of P_2, with the same value of the objective function.

$$P_1 : \min 2 - x \ s.\ t.\ x \geq 0 \ and\ x \leq 1$$

$$P_2 : -\max y \ s.\ t.\ y \geq -2 \ and\ y \leq -1$$

$$x \to y = x - 2$$

$$x = 0.5, y = -1.5$$

$$Obj.\ P_1 = Obj.\ P_2 = 1.5$$

Minimization or maximization

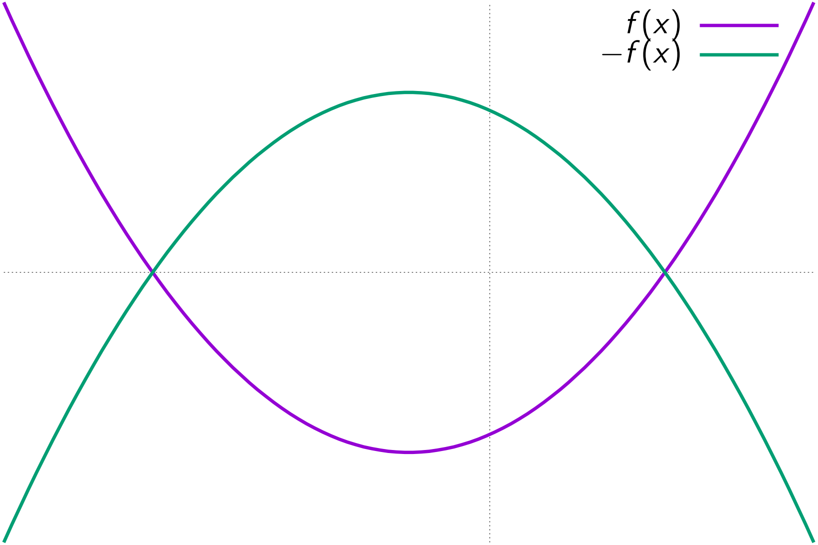
Write on the next slide

$$\min f(x) = -\max -f(x)$$

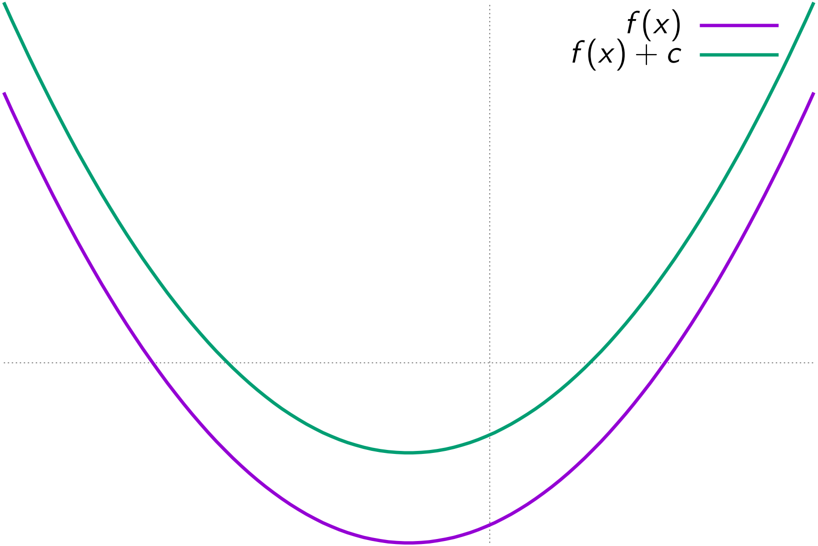
$$\operatorname{argmin} f(x) = \operatorname{argmax} -f(x)$$

and draw a line joining the two optima.

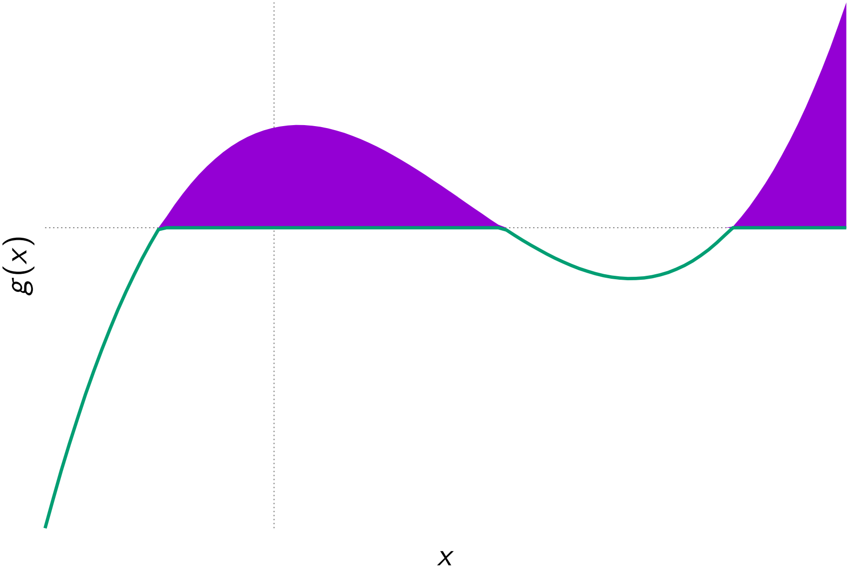
Minimization or maximization



Constant term



Inequality constraints

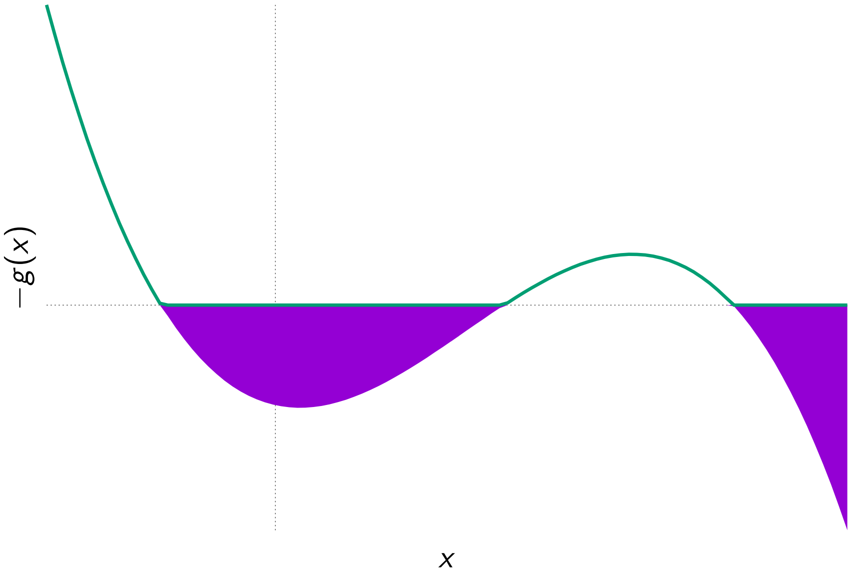


Inequality constraints

Write

$$g(x) >= 0 \Longleftarrow -g(x) <= 0$$

Inequality constraints



Equality and inequality constraints

$$g(x) = 0 \Longleftrightarrow \begin{cases} g(x) \leq 0 \\ g(x) \geq 0 \,. \end{cases}$$

Non negative variables

Some software impose only non negative variables
$x \in \mathbb{R}$, $x = x^+ - x^-$ *with* $x^+ \geq 0$, $x^- \geq 0$

Translation

Constraint $x \geq a$. *Change of variable:* $x = \tilde{x} + a$. *Constraint become* $\tilde{x} \geq 0$

Slack variables

Linear

$$g(x) \leq 0 \iff \begin{cases} g(x) + y = 0 \\ y \geq 0\,. \end{cases}$$

Non linear

$$g(x) \leq 0 \iff g(x) + z^2 = 0.$$

Summary

Several equivalent formulations of the same problem
We have seen valid transformations
Try it yourself on an example